Akash Dubey

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Education

Rutgers University Honors College, School of Arts and Sciences New Brunswick, NJ
Bachelor of Science in Computer Science and Mathematics *Anticipated May 2027*

- **GPA**: 3.95/4.0 | SAT: 1580
- **Relevant Coursework**: Stochastic Processes, Math Theory of Probability, Tensor Networks, Honors Calculus III & IV, Algorithms, Data Structures, Systems Programming, Computer Architecture.
- **Academic Programs & Research**: Physics Learning Theory Seminar (statistical mechanics in high-dimensional learning); Directed Research Reading Program on Symmetric Functions (representation theory for combinatorial optimization), Dean's List.
- **Activities**: Quantitative Finance Club, Road to Silicon Valley Program (Cohort 6), ESL Instructor, Pickleball.

Research

Kwan Lab, Rutgers Cancer Institute of New Jersey New Brunswick, NJ
Computational Biology Research Assistant (Advisor: Prof. Kelvin Kwan) *Jan 2026 – Present*

- Developed an AlphaGenome-based in silico deletion screen spanning 12 candidate sites, 3 deletion sizes (10, 40, and 80 kb), and 2 cell types (72 perturbation conditions) to quantify TAD-boundary sensitivity; identified a chr12 insulator with the strongest predicted insulation loss among 12 candidates.
- Built a 3D chromatin polymer simulator using overdamped Langevin dynamics to generate conformations from Hi-C contact matrices; implemented multi-scale TAD detection using insulation score and directionality index methods and validated the pipeline with 70+ automated tests.
- Configured and validated a reproducible DeepSTARR inference environment, resolving legacy TensorFlow/Keras dependencies in HPC to enable sequence-based prediction of developmental and housekeeping enhancer activity.

Rutgers Economics Labs New Brunswick, NJ
President (formerly Research Director, Team Lead) *Oct 2024 – Present*

- Leading quantitative research projects for partners including NJDOL, NJDEP, NJBPU, Federal HUD, U.S. Census Bureau; applied econometric and statistical models in Python/R to analyze large datasets and project economic trends.
- Architected an AI automated research pipeline RAIL (Rutgers Agentic Intelligence Labs) to collect, organize, analyze, and synthesize data to derive insights using data ontologies and coding agent workflows.
- Previously led a team of five researchers to evaluate the efficacy of the Urban Enterprise Zone program for the NJDCA, utilizing econometric methods in R and Python to analyze complex census datasets.

Algorithmic Robotics and Control Lab in Rutgers's Dept. of CS New Brunswick, NJ
Research Assistant (Advisor: Prof. Jingjin Yu) *Jan 2026 – Present*

- Implemented and validated CUDA versions of pRRTC and FCIT*, parallelizing tree/graph search and collision checking for high-dimensional ASAO robot motion planning across 500 benchmark problems.
- Developing a graph-based search policy (GaP) for dual-arm task-and-motion planning (TAMP), learning from successful and failed planner traces to rank symbolic action skeletons and predict geometric feasibility in Isaac Sim.

Engineering Experience

New York Life Insurance New York, NY
Software Engineer Intern, TDAV Team *May – Aug 2026*

- Created an AI text and voice agent for processing insurance claims for clients and 12,000 Customer Service Representatives using AWS Infrastructure with access to NYL services, knowledge, and forms.
- Constructed a company knowledge base from 13,000 documents compiled from Sharepoint, Public Site, Confluence, Jira, and the Intranet, bringing business logic into one place for AI and human users.

Scarlet Sync New Brunswick, NJ
Founder *Jan 2025 – Present*

- Architected a robust Python data ingestion pipeline using LLMs to reverse-engineer legacy university data, indexing over 2,500 classes and 500+ degree programs for real-time querying on a platform with thousands of users.

– Developed a production AI assistant with streaming tool-calling and grounded retrieval over schedules, degree plans, and profile context across Web, ChatGPT, Claude, and iOS.

Lykke San Francisco, CA
Founding Software Engineer *Sep 2025 – Present*

– Engineered adaptive RAG pipelines across MongoDB, Zilliz/Milvus, S3, Gemini Files API, and SSE streaming, including source inference, citation events, token-budgeting for large materials, and fallback reasoning paths.
– Created a distributed wiki-generation engine that crawls sources, synthesizes structured sections, embeds content for retrieval, and runs tenant-isolated background builds via SQS, MongoDB leases, Redis rate limiting, and AWS ECS.

Samaritan Scout Cranford, NJ
Full Stack Engineer Intern *May 2023 – Aug 2025*

– Developed an agentic scraping system leveraging OpenAI and Gemini structured outputs to autonomously extract data from over 100,000 websites, reducing manual data entry time by over 90%.
– Developed a full-stack search engine using React, TypeScript, and PostgreSQL to index 35,000+ records, implementing semantic search algorithms with hallucination filters to ensure accurate matching.

Selected Technical Projects

SLM Optimization & Post-Training (GSM8k) | *PyTorch, HPC*

– Fine-tuned small language models (SLMs) on the GSM8k math reasoning dataset to study the effects of different model sizes and architectures on math reasoning capabilities.
– Deployed models on the Rutgers High Performance Computing (Amarel) cluster, utilizing quantization and structural pruning to maximize inference throughput on constrained compute.

Knowledge Base and AI Agent Workflow Manager | *Data Ontology, Workflows, Coding Agents*

– Built KRAIL (pypi), a local-first, repo-backed knowledge runtime that structures raw notes into ontology-aware topic pages, markdown graphs, source ledgers, claim records, verification runs, and artifact lineage for evidence-grounded agent memory.
– Implemented MCP/CLI and agent infrastructure for auditable coding-agent workflows, including local runner dispatch, dry-run work orders, session logs, role prompts/checklists, workflow validation, and hybrid keyword-graph-vector retrieval.

Local Coworker: Native macOS Computer-Use Agent | *Swift, MCP, Gemini Vision* (private code)

– Built a native desktop agent capable of autonomous computer use by integrating Gemini Vision with macOS Accessibility APIs, AppleScript, and JXA, using MCP to orchestrate specialized sub-agents for multi-app workflows.

Awards & Skills

Awards: 8090 AI Top Coder (5th), Rutgers Shark Tank (2nd and 3rd), ASA Fall Data Challenge (Top 3), NSF I-Corps.
Languages: Python, C++, CUDA, R, Java, SQL, TypeScript, Bash, Git
Technologies: PyTorch, TensorFlow/Keras, HPC Clusters, Simulation (Langevin Dynamics), Statistics, Pandas, Scikit-learn, Transformers, AWS